Cardiovascular System (CARD) · NPLEX-P1-CARD-02

CV-2 Histology of the Heart and Blood Vessels

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-02-01-C01				
NPLEX-P1-CARD-02-01-C02				
NPLEX-P1-CARD-02-01-C03				
NPLEX-P1-CARD-02-01-C04				
NPLEX-P1-CARD-02-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C01-BD1

Draw the heart wall and the fibrous skeleton

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-BD1 Date: _____

Without looking at your notes, draw a cross section of the left ventricular wall from the cavity outward. Label the endocardium with its subendocardial layer, the myocardium, and the epicardium, and write beside each layer the tissue it is made of and what runs inside it. Write the vessel layer each heart layer corresponds to. Give the normal thickness of the atrial wall, the right ventricular wall, and the left ventricular wall. Then draw the heart in cross section at the level of the valves and lay out the fibrous skeleton: the four fibrous rings, the two fibrous trigones, and the membranous septum. Mark the thickest part, give its other name, and write which structure passes through it and where that structure runs next. Finish by writing the three layers of a valve leaflet with what each layer does, and name the layer that expands in myxomatous mitral valve disease.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C01-BD2

Wall stress, hypertrophy, and two failures of the skeleton

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-BD2 Date: _____

Write the relationship between wall stress, chamber pressure, chamber radius, and wall thickness. Using it, work three cases and show your numbers: a ventricle with a radius of 2.5 cm and a wall of 1.0 cm at a systolic pressure of 120 mmHg; the same ventricle when systolic pressure rises to 200 mmHg; and a ventricle with a radius of 3.5 cm and a wall of 0.8 cm at 120 mmHg. For each case write what the number means for the work each myocyte has to do. Then write how the wall remodels in pressure overload and in volume overload, where the new sarcomeres go in each, what happens to cavity size, and what the nuclei of hypertrophied myocytes look like under the microscope. Write one cost of thickening the wall. Finish with two failures of the fibrous skeleton: write what happens electrically when a strand of myocardium crosses a fibrous ring, including the three tracing findings that follow, and write why an abscess of the aortic valve ring in infective endocarditis can produce new conduction block.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C01-RT01

Rebuild why the subendocardium is injured first

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-RT01 Date: _____

Rebuild the six-step sequence that explains why the inner third of the ventricular wall is the first layer to become ischemic, one step per line and in order. Beside the step about systole, draw a small cross section of the wall and mark where intramural pressure is highest and which vessels are compressed. Beside the step about demand, write why wall stress is highest in the inner layers, using the relationship you already know. Then write the three situations named in the notes that tip supply below demand, write during which part of the cardiac cycle the inner wall is perfused, and write what the resulting infarct is called and why it can happen without an artery being completely blocked.

Rebuild from memory: Why the subendocardium is injured first

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

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What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-02-01-C01-CA01

Chest tightness on exertion in a man with a thick left ventricle

A 71-year-old man comes to the physician because of 6 months of tightness in the chest that comes on when he walks uphill and settles within a few minutes of stopping. Temperature is 36.8 C (98.2 F), pulse is 76/min, respirations are 16/min, and blood pressure is 128/80 mm Hg. A harsh systolic murmur is heard over the right upper chest and the carotid upstroke is slow and low in volume. Echocardiography shows a left ventricular wall thickness of 1.8 cm with a cavity radius of 2.4 cm, a peak systolic pressure in the left ventricle of 210 mm Hg, and vigorous contraction. Coronary angiography shows smooth epicardial coronary arteries with no narrowing.

1. Recognize. Name the pattern of hypertrophy present and the loading problem that produced it, and state where the new sarcomeres have been laid down.

2. Explain. Calculate wall stress for a radius of 2.4 cm at 210 mm Hg with a wall of 1.0 cm and again with a wall of 1.8 cm, then explain from the two numbers what the thickening achieved.

3. Differentiate. Write how this man's chest tightness differs in mechanism from chest tightness caused by a plaque narrowing an epicardial coronary artery, and name the finding here that separates the two.

4. Investigate. Write which layer of the ventricular wall is at greatest risk and which three features of a stained section of the wall you would look at to support that.

5. Decide. State the layer in which necrosis would appear first if his coronary perfusion fell during exertion, and name the type of infarct that results.

6. Justify. Justify that answer from how the coronary arteries enter the wall, what systole does to the inner vessels, and how wall stress is distributed from endocardium to epicardium.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C02-BD1

Draw a cardiac myocyte and its intercalated disc

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C02 Activity: NPLEX-P1-CARD-02-01-C02-BD1 Date: _____

Without looking at your notes, draw a working ventricular myocyte joined to its neighbour. Give its width and length in micrometres and label the nuclei, the perinuclear zone and what it contains, the mitochondria with the fraction of cell volume they occupy, and the capillary beside the cell. Write which fuel and which pathway supply nearly all of its ATP, how long contraction survives complete ischemia, and after how long injury becomes irreversible. Then draw the intercalated disc enlarged, marking the transverse portion and the lateral portion and which junctions sit in each. Build a three column table with the headings junction, structure, and function, and fill a row for the fascia adherens, the desmosome, and the gap junction, naming the proteins in each. Finish by drawing the T tubule and sarcoplasmic reticulum arrangement, naming it and stating where in the sarcomere it sits.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C02-BD2

When the junctions fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C02 Activity: NPLEX-P1-CARD-02-01-C02-BD2 Date: _____

Three hearts have a single junctional defect each. In the first, ventricular connexin 43 is greatly reduced but the desmosomes and fascia adherens are normal. In the second, a mutation in plakophilin 2 weakens the desmosomes. In the third, gap junctions are normal in number but have moved from the ends of the cells to the lateral borders, as happens at the edge of an old infarct. For each heart write whether the cells stay attached, what happens to the speed and direction of electrical spread, and what rhythm problem follows and why. For the second heart name the disease, write which chamber it affects most, what replaces the lost myocytes, and name the recessive form along with its hair and skin findings and the reason the skin is involved. Finish by writing why an adult heart cannot replace the myocytes it loses and what it does instead in overload and after an infarct.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-02-01-C02-CP01

Cardiac muscle against skeletal muscle

Fill in every cell of the table from memory, then open the notes and correct what you missed. In the last column write the demand on the heart that each cardiac feature serves.

Feature	Cardiac muscle	Skeletal muscle	What the cardiac version achieves
Cell shape and position of nuclei			
Coupling between cells			
T tubule size, position, and number of cisternae paired with it			
What triggers sarcoplasmic reticulum calcium release			
Whether extracellular calcium is needed for contraction			
Whether tetanus is possible			
How force is graded			
Capacity to regenerate after injury			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-02-01-C02-Q01

Sudden death in a young athlete with a fibrofatty right ventricle

A 19-year-old man collapses during a football match and is resuscitated from ventricular tachycardia. His father died suddenly at the age of 32. Examination shows woolly hair and thickened skin on the palms and soles. Cardiac imaging shows a dilated right ventricle with areas of the wall replaced by fat, and a biopsy specimen shows myocytes pulled apart from one another with fibrofatty tissue between them. Which of the following structures is most likely to be defective in this patient?

A. Connexon channels of the gap junction

B. Cadherin to actin linkage of the fascia adherens

C. Plakoglobin of the desmosome

D. Ryanodine receptor 2 of the sarcoplasmic reticulum

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C03-BD1

Purkinje fibers, working myocytes, and nodal cells

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-BD1 Date: _____

Without looking at your notes, write where Purkinje fibers lie in the ventricular wall, what they are continuous with, and how a stained section tells you a cell is one: its size, the colour and content of its cytoplasm, the stain it reacts with, where its myofibrils sit, and what surrounds its nucleus. Write the three features that make it the fastest conducting cell in the heart and give its conduction velocity. Then build a table from memory with one column each for the Purkinje fiber, the working ventricular myocyte, and the nodal cell, and rows for size and staining, myofibrils, gap junctions, conduction velocity, and main role. Finish by writing the rate at which Purkinje fibers pace, what that rhythm looks like on a tracing, and the unrelated cell with the same name that is a common matching trap.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C03-BD2

Read two sections and follow the consequences

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-BD2 Date: _____

A section taken just beneath the endothelium of the left ventricle shows large pale cells that stain strongly with periodic acid Schiff, have few myofibrils at the edge of the cell, and have a clear halo around the nucleus. A section of the right atrial wall shows striated cells with small membrane bound dense granules beside the nucleus and Golgi apparatus. Name each cell and write what it does. For the first, write why its stored glycogen matters when the inner wall becomes ischemic, and what the surviving cells at the inner edge of an infarct can start. For the second, name the hormone in the granules, write the stimulus that releases it, and write its receptor, its second messenger, and its four actions on the kidney and the vessels. Then write which chamber makes the related peptide that is measured in blood, why that one is not stored in prominent granules, and what its plasma level does in heart failure and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C03-RT01

Rebuild the atrial stretch feedback loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-RT01 Date: _____

Rebuild the five-step loop that runs from a rise in blood volume back to a fall in atrial stretch, one step per line and in order, then draw it as a closed loop with arrows. Beside the release step write which cells release the hormone, where in the cell it was stored, and which atrium holds the most of these granules. Beside the receptor step write the receptor name and the enzyme domain it carries, and write which second messenger that produces. Beside the kidney step write what happens to each of the two glomerular arterioles and what that does to filtration. Then write in one line what kind of feedback this loop is and how you can tell from the last step.

Rebuild from memory: Atrial stretch and natriuretic peptide feedback

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C04-BD1

Draw the wall of every vessel class

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C04 Activity: NPLEX-P1-CARD-02-01-C04-BD1 Date: _____

Without looking at your notes, draw cross sections of an elastic artery, a muscular artery, an arteriole, a capillary, a postcapillary venule, a medium vein, and a large vein, side by side and roughly to scale. Label the three layers on each where they exist, and write beside each drawing the feature that identifies it down the microscope and its main job. Give the number of elastic lamellae in the aortic media, the number of smooth muscle layers in a muscular artery media and in an arteriole, the diameter of a capillary, and the diameter below which a vessel counts as an arteriole. Mark where the internal elastic lamina is most distinct and where the vasa vasorum run, and write what the vasa vasorum are for and which vessel type has more of them. Finish by writing where valves are found and the three places they are absent, and what fraction of blood volume the veins hold.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C04-BD2

Which layer is damaged, and what the pressures do

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C04 Activity: NPLEX-P1-CARD-02-01-C04-BD2 Date: _____

For each of these, name the vessel layer or component that is damaged and write how the damage explains the finding: an older patient with a high systolic pressure and a low diastolic pressure; a tall young patient with a defect in fibrillin 1 who has a dilated aortic root and tearing chest pain; a patient with tertiary syphilis who has an aneurysm of the ascending aorta and an intima described as tree bark; a patient whose muscular arteries show pipestem calcification on an X ray with no narrowing of the lumen; and a patient with long standing hypertension and diabetes whose renal arterioles are thickened by pink material. Then write the relationship between resistance and vessel radius, and work out roughly how much resistance rises when an arteriole narrows by 20%. Write the pressure at the start of the arterioles and the pressure at their end, and explain why most of the pressure drop happens there. Finish by writing what the elastic recoil of the aorta contributes in diastole and what a loss of elastin does to systolic pressure, diastolic pressure, and pulse pressure.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-02-01-C04-Q02

Capillary type and what can leave the vessel

A researcher examines electron micrographs of capillaries from four tissues. One micrograph shows a wide lumen, large gaps between endothelial cells, and a basal lamina that is present in some places and absent in others. Which of the following is most likely to pass through the wall of the vessel in this micrograph?

A. Whole red and white blood cells

B. Water and dissolved gases only

C. Small peptide hormones but not albumin

D. Nothing larger than glucose, because tight junctions seal the wall

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C05-BD1

Build the endothelial product table and the nitric oxide pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-BD1 Date: _____

Without looking at your notes, build a three column table with the headings product, mechanism, and effect, and fill a row for nitric oxide, prostacyclin, endothelin-1, von Willebrand factor, P-selectin, thrombomodulin, heparan sulfate, and tissue plasminogen activator. Then write the nitric oxide pathway in full: name the enzyme, its substrate and cofactors, both products, and the four things nitric oxide does besides relaxing smooth muscle. Write what destroys nitric oxide and how long it survives, and what that means for how far it acts. Draw the storage granule that holds von Willebrand factor, give its shape and name, list what is in it and in its membrane, and write how quickly it is released and by which four stimuli. Finish by writing the two jobs von Willebrand factor does, the enzyme that trims it, and the disease caused when that enzyme is missing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C05-BD2

Two vessels, one drug, and the same receptor

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-BD2 Date: _____

Acetylcholine is applied to two arterial rings. The first has intact endothelium and relaxes. The second has had its endothelium rubbed away and contracts. Write the receptor acetylcholine binds in each ring, the cell it acts on, the signalling route it uses, and why the two rings move in opposite directions. Then write what happens when acetylcholine is infused into a human coronary artery that has atherosclerosis, and what that response reveals. Next write what rising shear stress does to an artery and through which molecule. Finish with aspirin: write which enzyme it modifies and how, why the platelet cannot recover the enzyme while the endothelial cell can, and what that difference does to the balance between thromboxane A2 and prostacyclin at a low dose.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C05-RT01

Rebuild the nitric oxide relaxation pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-RT01 Date: _____

Rebuild the six-step pathway from a stimulus at the endothelial surface to a dilated vessel, one step per line and in order. Beside the first step write the three stimuli given in the notes. Beside the enzyme step write what activates the enzyme. Beside the cyclase step write which cyclase it is, which nucleotide it makes, and which kinase that nucleotide activates. Beside the last two steps write what happens to calcium inside the smooth muscle cell and which enzyme acting on myosin is switched on. Underneath, write what the same signal does to platelets, and write which other hormone in this chapter raises the same second messenger and through which receptor.

Rebuild from memory: Nitric oxide and vascular relaxation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-02-01-C05-PR01

Predict what happens to an endothelium under oxidative stress

A 58-year-old man has smoked for 30 years and has type 2 diabetes, untreated hypertension, and a high level of oxidized LDL. Predict each item below, write the direction of the change, and give the molecule or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the amount of superoxide in the endothelial cell		
What happens to the amount of nitric oxide available to the smooth muscle		
What happens to the enzyme that makes nitric oxide once tetrahydrobiopterin runs low, and what it starts producing instead		
What happens to endothelin-1 and what that does to vessel diameter		
What happens to VCAM-1, ICAM-1, and the selectins on the endothelial surface, and what that does to monocytes		
What happens to tissue factor, plasminogen activator inhibitor 1, and thrombomodulin, and what that does to the surface of the vessel		
What this vessel does when acetylcholine is infused into it, compared with a healthy vessel		
Which parts of the arterial tree the resulting lesions appear in first, and what about the flow there explains it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____